

1.30 evaluate $KL(P||q)$ where $p = \mathcal{N}(x|4\sigma^2)$
 $q = \mathcal{N}(x|m, \sigma^2)$

$$KL(P||q) = E_p \left[\ln \frac{p}{q} \right] = \underbrace{E_p [\ln p]}_{(1)} - \underbrace{E_p [\ln q]}_{(2)} \quad (\text{linearity})$$

$$\textcircled{1} \quad E_p [\ln p] = \int (2\pi\sigma^2)^{-1/2} \cdot e^{-\frac{(x-\mu)^2}{2\sigma^2}} \cdot \ln \left((2\pi\sigma^2)^{-1/2} \cdot e^{-\frac{(x-\mu)^2}{2\sigma^2}} \right) dx$$

$$= (2\pi\sigma^2)^{-1/2} \int \ln \left((2\pi\sigma^2)^{-1/2} \right) e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$$

$$+ (2\pi\sigma^2)^{-1/2} \int \frac{-(x-\mu)^2}{\sigma^2} \cdot e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$$

$$= \ln \left((2\pi\sigma^2)^{-1/2} \right) \int (2\pi\sigma^2)^{-1/2} \cdot e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$$

$$- \frac{1}{2\sigma^2} \int (x-\mu)^2 \cdot (2\pi\sigma^2)^{-1/2} \cdot e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$$

σ^2

$$= -\frac{1}{2} (1 + \ln(2\pi\sigma^2))$$

$$\textcircled{2} E_p[\ln q]$$

$$= \int (2\pi\sigma^2)^{-1/2} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \ln \left((2\pi\sigma^2)^{-1/2} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \right) dx$$

$$= \ln(2\pi\sigma^2)^{-1/2} \int (2\pi\sigma^2)^{-1/2} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$$

$$- \frac{1}{2\sigma^2} \int (x-\mu)^2 \cdot (2\pi\sigma^2)^{-1/2} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$$

$$= \ln(2\pi\sigma^2)^{-1/2} - \frac{1}{2\sigma^2} \int (x-\mu + \mu - \mu)^2 \cdot (2\pi\sigma^2)^{-1/2} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$$

$$= \ln(2\pi\sigma^2)^{-1/2} - \frac{1}{2\sigma^2} \int ((x-\mu)^2 + 2(\mu-\mu)(x-\mu) + (\mu-\mu)^2)$$

$$(2\pi\sigma^2)^{-1/2} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$$

$$2(\mu-\mu)x + 2\mu\mu - 2\mu^2 \\ + \mu^2 - 2\mu\mu + \mu^2 \\ \Rightarrow \mu^2 - \mu^2$$

$$= \ln(2\pi\sigma^2)^{-1/2} - \frac{1}{2\sigma^2} \left(\int (x-\mu)^2 \cdot (2\pi\sigma^2)^{-1/2} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx + \right.$$

$$\left. 2(\mu-\mu) \int x \cdot (2\pi\sigma^2)^{-1/2} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx + (\mu^2 - \mu^2) \int (2\pi\sigma^2)^{-1/2} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx \right)$$

$$= \ln(2\pi s^2)^{-1/2} - \frac{1}{2s^2} (\sigma^2 + 2(N-m)\mu + m^2/\mu^2)$$

$$= -\frac{1}{2} \ln(2\pi s^2) - \frac{1}{2s^2} (\sigma^2 + \mu^2 - 2m\mu + m^2)$$

$$= -\frac{1}{2} \left(\ln(2\pi s^2) + \frac{\sigma^2 + (N-m)^2}{2s^2} \right)$$

$$(\text{if } s^2 = \sigma^2, \mu = m, E_p[\ln q] = E_p[\ln p])$$

$$\therefore KL(P||Q) = E_p[\ln p] - E_p[\ln q]$$

$$= -\frac{1}{2} (\ln(2\pi \sigma^2) + 1 - \ln(2\pi s^2) - \frac{\sigma^2 + (N-m)^2}{2s^2})$$

$$= \frac{1}{2} \left(\ln\left(\frac{s^2}{\sigma^2}\right) + \frac{\sigma^2 + (N-m)^2}{2s^2} - 1 \right)$$
